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Self-Monitoring for High School Students With Disabilities: A Cross-Categorical Investigation of I-Connect

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Abstract

Self-monitoring interventions are well supported within the empirical literature as improving classroom engagement for students with disabilities. However, studies implementing self-monitoring interventions in high school settings are rarely conducted despite their potential to improve student academic and behavioral outcomes. In an investigation of an unobtrusive, self-monitoring application loaded on a handheld device, classroom engagement and perceived academic benefits were assessed in a withdrawal design for three high school students with different disabilities (specific learning disability, autism, and intellectual disability) in varied instructional arrangements. Direct observation data supported the intervention as effective in improving classroom engagement for all three students during intervention and maintenance phases.

Keywords

self-management, classroom, intervention(s), adolescent, positive behavior support(s)

High school students with disabilities represent a critically underserved population for intervention research. Despite advances within school-wide positive behavior support (SWPBIS) literature (Bradshaw, Mitchell, & Leaf, 2010; Lewis & Sugai, 1999; Sugai & Horner, 2009), the majority of studies investigating universal and secondary behavioral interventions have been conducted within elementary classrooms. Structural characteristics common to high school settings may contribute to the lack of empirical research. First, high school teachers may feel unequipped to intervene, particularly for challenging behavior (Baker, 2005). Second, the number of classes, compartmentalization of departments, and administrative structures inherent to high schools may require adjustments to interventions and programs (Flannery, Sugai, & Anderson, 2009). Third, support provided to high school students must be timely and effective but avoid stigma that may be experienced during the transition from universal to secondary interventions (Rapee et al., 2006). Ideally, interventions chosen for high school settings would capitalize on existing school and classroom resources, allow for selective and portable implementation for students who did not respond to universal intervention, and avoid any potential stigma that may be associated with obtrusive intervention delivery in the presence of classroom peers.

Self-management interventions represent a potential solution for schools hoping to meet all three criteria. Broadly defined, self-management is the application of self-directed behavior strategies to change behaviors in a desired way (Cooper, Heron, & Heward, 2007). Although self-management interventions are typically multicomponent and encompass a large number of interventions (Fantuzzo & Polite, 1990), self-monitoring is the most commonly incorporated element typically implemented in self-management interventions. Self-monitoring is a self-directed behavior strategy that consists of observing and recording one's own behavior (Mace, Belfiore, & Hutchinson, 2001).

Several studies have shown self-monitoring to improve academic performance in middle and high school settings including (a) reading comprehension (Alfassi, 1998; Kozminsky & Kozminsky, 2001), (b) accuracy in reading and math assignments (Cancio, West, & Young, 2004; Shimabukuro, Prater, Jenkins, & Edelen-Smith, 1999), and (c) increased productivity in writing (Trevino-Maack, Kamps, & Wills, 2014). Self-monitoring interventions have been shown to improve behaviors that contribute to progressing in school such as lesson and task completion

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(Foley & Epstein, 1993; Koegel, Harrower, & Koegel, 1999; Seabaugh & Schumaker, 1994). Others have successfully taught middle school students to self-monitor preparedness, engagement, and homework completion (Chafouleas, Sanetti, Jaffery, & Fallon, 2012; Wood, Murdock, & Cronin, 2002). Furthermore, middle and high school studies incorporating self-monitoring have shown improved on-task behavior (Blick & Test, 1987; Faul, Stepensky, & Simonson, 2011; Lloyd, Bateman, Landrum, & Hallahan, 1989; Mitchem, Young, West, & Benyo, 2001), reductions in stereotypic behaviors (Crutchfield, Mason, Chambers, Wills, & Mason, 2015), and decreased disruptive behaviors (Peterson, Young, Salzberg, West, & Hill, 2006; Smith & Sugai, 2000; Stage & Quiroz, 1997.

Despite the research support for self-monitoring interventions across settings and behaviors, gaps exist within the literature. Technological advances have only recently been incorporated into school-based investigation of self-monitoring despite the rapid proliferation of handheld devices. The use of handheld mobile technologies allows for efficient data collection and unobtrusive prompting in school settings; however, only rarely are higher tech strategies selected over paper-based systems. When technological advances are incorporated, positive results have been found, although the capabilities of the devices to record, store, and transfer self-monitoring data are rarely realized. More commonly, participants have recorded their self-monitoring data on a paper-based form after prompting by the device. Recent studies using similar self-monitoring procedures have incorporated handheld computers (Cihak, Wright, & Ayres, 2010), an iPad touch (Blood, Johnson, Ridenour, & Simmons, 2011), or a cell phone (Quillivan, Skinner, Hawthorn, Whited, & Ballard, 2011) as the prompting device.

One of the few studies to incorporate the technological component of the self-monitoring intervention both to prompt and record participant responses was conducted by Gulchak (2008). The study used a mobile handheld computer (Palm Zire 72) to prompt self-monitoring for an 8-year old with a diagnosed emotional and behavioral disorder. During independent reading periods in a self-contained classroom, the device prompted on-task appraisals with a scheduled chime at 10-min intervals. The intervention resulted in improved on-task behavior (average of 64% in baseline to 90% during intervention). The author stated that the device was highly reinforcing for the student, an observation supported by the high fidelity of participant use observed within the study (100%). Extending the work done by Gulchak, Wills and Mason (2014) conducted a study of self-monitoring using an application loaded on a handheld device (Samsung Galaxy Player 5.0) for two students with disabilities within an inclusive high school science classroom. Using unobtrusive (flashing) prompts delivered at 5-min intervals, the study resulted in improved on-task behaviors for both participants and collateral improvements in frequency of disruptive behaviors for one student. Although the study supported the use of technologically prompted self-monitoring in the classroom setting, additional research is needed to determine whether the intervention is more or less efficacious for high school students in different instructional arrangements being served for different categorical disabilities.

Purpose

The purpose of the current study was to assess the effectiveness of an unobtrusive, technologically delivered self-monitoring intervention to address the following research questions:

Research Question 1: Does the intervention demonstrate efficacy in improving on-task behavior for participating high school students with disabilities?
Research Question 2: Is the intervention considered acceptable and efficacious by participating classroom teachers and students with disabilities?

Method

Participants

Participant recruitment was conducted after district approval was granted and institutional review board (IRB) approval was received. Student participants were nominated for study inclusion by school-based intervention teams based on two criteria: (a) presence of a diagnosed disability as evidenced by the individualized educational plans and (b) teacher-referred difficulties with maintaining on-task behavior in the classroom that was contributing to academic difficulties with the possibility of grade failure to insufficient lesson completion. Once student participants were identified, parental consent and student assent were obtained, and brief observations were then conducted to confirm high rates of off-task behavior. Participant 1, Keith, was a 17-year-old, White male student in the 11th grade receiving special education services for a specific learning disability in the areas of reading comprehension, written expression, and math calculation. Keith was observed in his ecology class throughout the entire length of the study. Participant 2, Brad, was a 17-year-old, White male student in the 11th grade receiving special education services under the diagnostic criteria for autism. Brad was observed in his technical English class the entire duration of the study. Participant 3, Miranda, was a 15-year-old, White female student in the 9th grade receiving special education services under the diagnostic criteria for intellectual disability. All observations for Miranda were conducted in a self-contained classroom. For a summary of participant characteristics, see Table 1.

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Table 1. Participant Demographics.

Participant	Age	Grade	Sex	Percentage of time in GenEd	Diagnosis	Setting for study
Keith	17	11th	Male	67	Specific learning disability	Ecology
Brad	17	11th	Male	88	Autism	Technical English III
Miranda	15	9th	Female	48	Intellectual disability	Self-contained setting

Setting

The study was conducted in a rural, public high school in Western Missouri. According to information provided by the state's Department of Education, the district served approximately 1,000 students and approximately 49% of the district's predominately White students were considered to be economically disadvantaged. The high school's enrollment was approximately 330 students and reflected the ethnic and socioeconomic composition of the broader district. The school served 35 of the 39 students receiving special education services in general education classrooms for more than two thirds of instructional time, whereas the remaining 4 students served primarily in a resource room or a self-contained classroom. Both general education (ecology and technical English III) and self-contained settings were used for baseline and intervention data collection. Each class period lasted approximately 50 min on average. During the two semesters in which this study was conducted, the participating high school was not implementing a Positive Behavior School-Wide Program at the high school, although they had instructed students in school-wide expectations (posted throughout the school). Infractions and disciplinary consequences were clearly documented, and student and school-wide disciplinary data were incorporated in team problem-solving sessions.

Materials and Equipment

The materials selected for this study included a Gymboss interval timer, a reinforcer menu, and a Samsung Galaxy Player 5.0 tablet loaded with a self-monitoring application (I-Connect). A Gymboss interval timer was used for all direct observations of the dependent variables. The timer was set to vibrate every 15 s throughout the duration of the 30-min observation periods. During all observations, the observer wore the timer on the belt or in the pocket. A reinforcement menu was created for each student using a word processor software program that listed 10 potential reinforcement items embedded with picture icons.

During the intervention phases, each participant was given a Samsung Galaxy Player 5.0 that had the I-Connect application installed. The Samsung Galaxy Player 5.0 is a handheld tablet with a 3 × 5 inch touch screen. The tablet had Wi-Fi capability and allowed the student to monitor his or her behavior utilizing the I-Connect application but did not possess call or text capabilities. Fixed intervals were selected based on the typical duration of on-task behavior during baseline, with Keith and Brad receiving 1-min prompts and Miranda receiving 30-s prompts. When the interval time was over, the screen flashed "Are you on-task?" as the prompted question. Although the I-Connect self-monitoring application included additional monitoring capabilities (e.g., attendance, homework completion), only the self-monitoring of on-task behaviors was used in the present study.

Research Design

This study used an ABAB single-subject withdrawal design for each participant (Kazdin, 1982; Kennedy, 2005). An ABAB single-subject withdrawal design allowed the researcher to demonstrate a functional relationship between the intervention and the targeted behavior by using a withdrawal or return to baseline phase. On-task behavior was identified as the targeted behavior for this study. If a functional relationship existed, the amount of on-task behavior would increase during the intervention phase while the participant used the I-Connect application and decrease during the return to baseline phase when the I-Connect application was withdrawn.

Measurement

On-task behaviors. On-task behavior was measured using direct observations. On-task was included the following: attending to the materials or teacher, sitting up at their desk, and completing work during independent tasks. Data were recorded using a 15-s momentary time-sampling recording system during 30-min observation sessions. For each session, 120 intervals were recorded. A Gym-Boss interval timer prompted observers to record student behavior with a vibration prompt at the end of each interval. If the student demonstrated off-task behavior at the end of the interval, the entire interval was scored as off-task. If the student was on-task at the end of the interval, the interval was recorded as on-task. Failures to respond to the I-Connect application were included in the number of intervals recorded as off-task by the direct observer and are not represented in the percentage of on-task intervals recorded by the participant.

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A second observer was trained to collect data for the purposes of interobserver agreement (IOA). The second observer met with the primary observer to review operational definitions and procedures for data collection. The two observers watched video recordings of observational sessions and recorded momentary time-sampling data. When the two observers obtained 90% agreement, then IOA was collected for all participants across phases. Throughout the study, the second observer watched video recordings of the observational sessions to conduct IOA. IOA was collected for 20% of sessions per phase of the study for each participant. IOA was calculated using the point-by-point agreement method: the number of agreements divided by the number of agreements + the number of disagreements × 100. Mean point-by-point agreements ranged from 91% to 100% across all phases and participants.

Social validity. To assess the social validity and consumer satisfaction of the I-Connect self-monitoring application, the participating teachers and students completed Consumer Satisfaction Rating scales to indicate to what extent they thought the I-Connect application was effective in increasing on-task behavior and classroom performance at the conclusion of the study. Two separate rating scales were developed and administered. A teacher rating scale provided the teachers the opportunity to rate the following: student productivity in class, assignment completion, grade, and accuracy of work using a 5-point Likert-type scale with 1 being *no improvement noted* to 5 being *great improvement noted*. The participating students also completed a 5-point Likert-type rating scale that required them to rate their classroom performance in the areas of work completion, grade, performance on class work, and attention during lectures and class work.

Procedures

Baseline. During baseline, the participants were observed in their high school classroom setting without any behavioral support strategies in place. Keith was observed during lectures, classroom assignments, tests, and group activities in his ecology class. Brad was observed during reading activities, group discussions, individual assignments, and classroom tests in his technical English class. Miranda was observed during her self-contained reading class while she worked on individualized reading assignments. The researcher did not adjust any element of the classroom setting during the baseline phase.

I-Connect training. Training was spread out over three sessions on separate days. Total training time was approximately 1 hr. Three training sessions were provided to each participant in a 1:1 setting with the first author. During Training Session 1, the first author used direct instruction and modeling to demonstrate what on-task behavior looked like. Each participant was provided with examples of on-task behavior (e.g., looking at your work, looking at the teacher while you listen to the teacher, writing the answers to your assignment) and non-examples of on-task behavior (e.g., looking around the classroom, talking with your peers, getting out of your seat). After the discussion about examples and non-examples, the researcher and the participant watched video recorded sessions from the baseline observations. The first author would pause the video and ask the student whether they were on-task at that time. The participant had to correctly identify on-task behavior for five consecutive trials before the training session would conclude.

During Session 2, the participants were taught how to use the I-Connect application and monitor their behavior. First, they were instructed how to turn it on and navigate to the I-Connect self-monitoring program. The first author demonstrated the skill, and then allowed the participant to try while providing them with verbal prompts. After the participant accurately demonstrated turning on the device and starting the application (with no more than two verbal prompts), the participants and the first author reviewed more baseline session video recordings. During this training session, the participants monitored their behaviors every interval just as they would during an intervention session. The first author monitored student responses and calculated their accuracy with recording. If the student recorded their behavior accurately, praise was provided; however, if they did not, the video was paused and the first author provided error correction. After the participant accurately responded to the application for a minimum of 10 consecutive trials, the training session ended. Following the training, the participant used the application during a mock intervention session. During this time, the researcher observed as the participant recorded his or her behavior for a 30-min session using the I-Connect application. The data from this observation would be used during the third training session.

On Training Day 3, the participant first reviewed the data collected at the end of the second training day. The participants were provided feedback based on their performance during the observation and the accuracy of their self-monitoring. The participants were then assessed as they used the application during another mock intervention session. The participants had to demonstrate their ability to turn the device on and accurately monitor their behavior with the I-Connect application for at least five trials during this session. After meeting the criterion, the participants began intervention the following class period.

Intervention. After each participant received training on how to use the I-Connect application, intervention procedures began. Two components were included during intervention phases: self-monitoring using the I-Connect application and reinforcement. During intervention phases,

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each participant had the responsibility of picking up the I-Connect application from the resource room before going to the class where the intervention was implemented. It was the participants' responsibility to turn the device on and begin monitoring their on-task behavior. During observations, the observer was available and provided assistance to the participants if there were any technical problems with the application (e.g., difficulty connecting to the wireless Internet). Other than providing technical support, the observer and the participants did not engage in other interactions. All three participants monitored their on-task behavior using the I-Connect application. The two participants in the general education setting monitored their on-task behavior at 1-min intervals, and the one participant, Miranda, in the self-contained setting monitored her on-task behavior at 30-s intervals. While self-monitoring, the participants were prompted by the I-Connect application to answer the question, "Am I on-task?" when the screen flashed to catch their attention and then displayed the question with the choices of "yes" or "no."

Prior to each intervention session, the participants selected an item from their individual reinforcement menus if they met the dual reinforcement criteria. To obtain the reinforcer, the participant had to accurately record their behavior by agreeing with the researcher a minimum 80% of intervals recorded, and the participant had to be on-task for a minimum of 80% of the time based on the momentary time-sampling data obtained by the researcher. At the end of the session, the observer would review the participants' I-Connect data online, calculate matching between the observer and participant, and discuss the results with the participant. If the participant did not meet the dual criteria, no reinforcement was provided.

Return to baseline. Return to baseline (A2) was the third phase during which the participants did not use the I-Connect self-monitoring application to manage their on-task behavior. The participants were made aware that the device and application would be unavailable for a few sessions and it would be returned at a later time.

Return to intervention. Intervention was reintroduced after a trend had occurred during the return to baseline phase with at least five sessions of data collected that indicated that a functional relationship existed between the use of the I-Connect application for self-monitoring and the increase of on-task behavior. Intervention data were collected to the end of the study for at least five sessions. Additional sessions might have been collected if the participant's self-monitoring data did not match the observer's data to at least 80% agreements.

Maintenance. To determine the effectiveness of the intervention over a period of time, maintenance probes (2 weeks and 4 weeks) were administered during the month following the study (Kazdin, 1982). During the maintenance phases, the students were allowed to use the application on a daily basis, and they were observed during one session 2 weeks after the final session of return to intervention (B2) and during one session 4 weeks after the final session of return to intervention (B2). These sessions were collected to determine whether the behavior was maintained when the contingencies for behavior stopped. Maintenance probes were impromptu observations of the participants' behavior using the I-Connect self-monitoring intervention during the same content area but at times they were not aware the sessions were going to occur. Matching data were collected during maintenance probes; however, if the participant's self-monitoring data did not match the observer's data to at least 80%, a re-teaching element was not included and additional sessions were not collected.

Results

The data from this study suggested that use of the I-Connect application resulted in improved on-task behavior for all three participants. During each intervention (B1 & B2) phases, there was an increase in on-task behavior for all three participants when using the application to self-monitor. On-task behavior decreased during the return to baseline (A2) phase, supporting a functional relationship between the use of the I-Connect self-monitoring application and the increase in on-task behaviors for each participant.

Participant 1: Keith

(A1) Baseline. Baseline data were collected for Keith for a total of six sessions. An average of Keith's on-task behavior was 45% with a range of 8% to 64% (see Figure 1).

(B1) Intervention–implementation of the I-Connect application. During the first phase of intervention, Keith was observed for a total of five sessions. During this phase, Keith's average on-task behavior was 92.2% with a range of average on-task behavior from 83% to 98%. During intervention phase, self-monitoring data were recorded using the I-Connect application. Reported on-task average was 96.6% per 30-min observation with a range of 93% to 100%.

(A2) Return to baseline. During return to baseline (A2), data were recorded a total of five data sessions. Keith was not allowed access to the I-Connect self-monitoring application or reinforcement during this phase. Keith was on-task an average of 66% of the time with a range of 58% to 78% of the time. This was a decrease in the amount of time on-task when compared with his average of 92.2% of the time on-task during the intervention phase.

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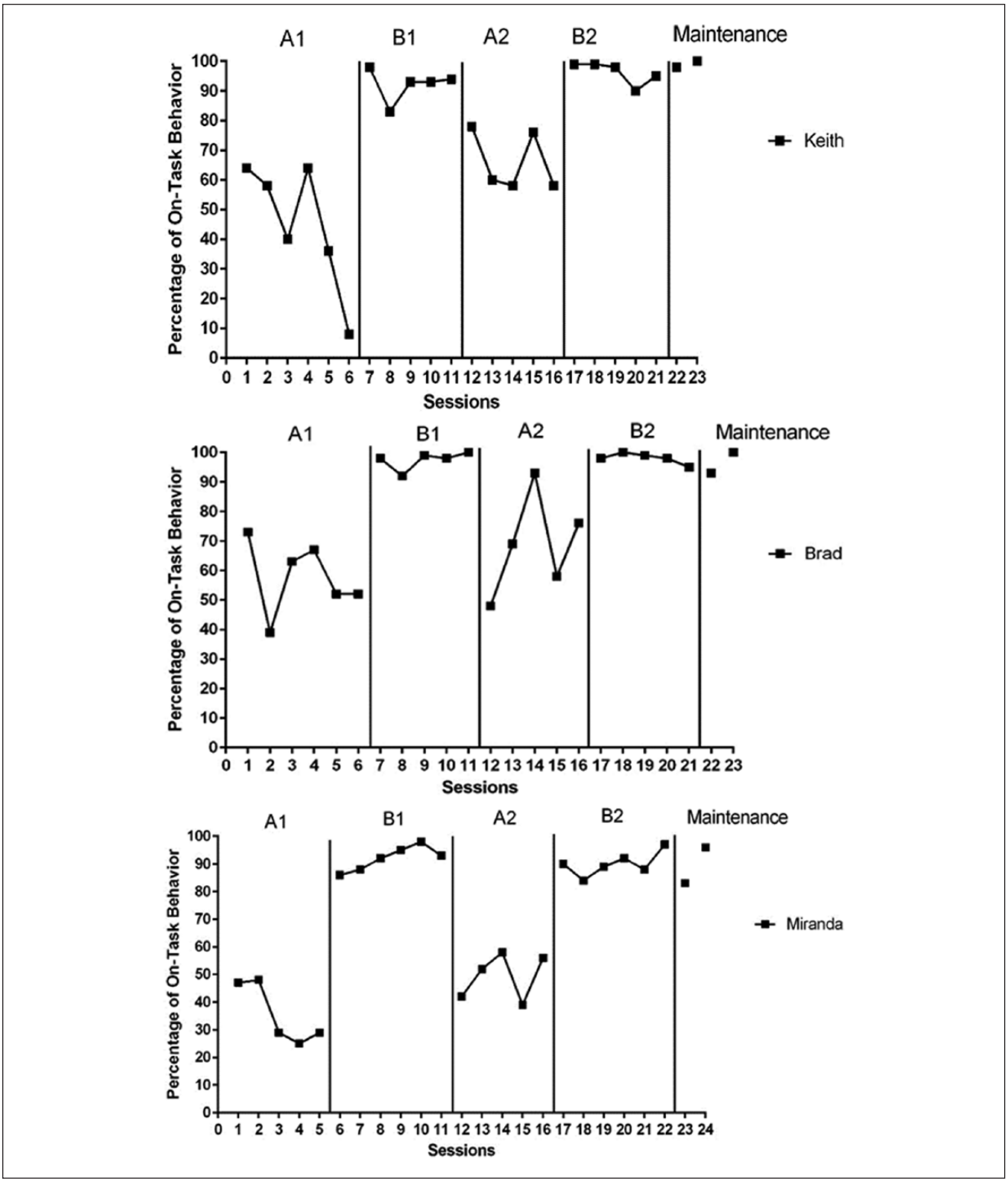


Figure 1. Participant results for on-task behavior.

(B2) Return to intervention. After five data sessions in A2, intervention was reintroduced to Keith for five sessions. During return to intervention phase, Keith was on-task an average of 96.2% of the time with a range of 90% to 99%. During return to intervention, the participant recorded his own behavior using the I-Connect application, and he noted

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that he was as on-task an average of 97.4% of reporting intervals with a range from 93% to 100%.

Maintenance probe results. Maintenance probes were administered at 2 weeks and 4 weeks after the last session of return to intervention (B2) phase. Keith was on-task 98% of the time and he answered 96% of the questions correctly. He recorded as on-task 100% of the intervals and matched the observer 100% of the time. During the 4-week maintenance probe, he was on-task 100% of the time and recorded his behavior as on-task 91% of the time, matching the observer for 91% of intervals.

Participant 2: Brad

(A1) Baseline. Baseline data were collected for a total of six sessions. The participant was observed to be on-task an average of 58% of the time with a range of 39% to 73% (see Figure 1).

(B1) Intervention–implementation of the I-Connect application. During Intervention (B1), observational data were collected for a total of five sessions. During intervention, Brad engaged in on-task behavior an average of 97.4% of the time with a range of on-task behavior of 92% to 100%. He recorded himself as on-task 100% of intervals across all sessions.

(A2) Return to baseline. During return to baseline (A2), the participant was observed a total of five data sessions; Brad's on-task behavior decreased to an average of 68.8% of the time with a range of 48% to 93% of the time on-task.

(B2) Return to intervention. The participant was observed for five data sessions during return to intervention phase. During these five sessions, Brad was on-task an average of 98% of the time during return to intervention with a range of 95% to 100%. The participant recorded himself as on-task an average of 97.4% of intervals with a range from 93% to 100%.

Maintenance probe results. Maintenance probes were administered 2 weeks and 4 weeks after the last session of return to intervention (B2) phase. The observed on-task behavior was on-task 93% of the time. And recorded himself as on-task 97% of the time. During the observation/I-Connect comparison, the observer and participant agreed 93% of intervals. During the 4-week maintenance probe, Brad was recorded as on-task for 100% of intervals. The observer and the participant agreed for 100% of intervals during the maintenance probe.

Participant 3: Miranda

(A1) Baseline. Miranda was observed a total of five sessions during Baseline (A1) and was on-task an average of 35.6% of the time with a range from 25% to 48% (see Figure 1).

(B1) Intervention–implementation of the I-Connect application. Once baseline data were collected, Miranda was trained to use the I-Connect application. Intervention data were collected for six sessions. During intervention, Miranda increased the amount of time on-task from an average of 35.6% (during baseline) to an average of 92%. The amount of time on-task ranged from 86% to 98% across sessions. During intervention she recorded her behavior as on-task an average of 92% (range = 86%–98%) of the time. During Session 5, Miranda failed to match the observer on at least 80% of intervals. This required that the intervention be extended for one session to provide the participant with additional instruction on use of the I-Connect application and the importance of accurately recording her behavior. During Session 6, the participant recorded her behavior accurately and matched the observer for 86.11% of intervals.

(A2) Return to baseline. During this phase, the observer recorded Miranda's behavior as on-task an average of 49.4% of the time, a decrease from the average 92% on-task during intervention. On-task behavior during return to baseline ranged from 39% to 58%.

(B2) Return to intervention. Return to intervention data were collected a total of six sessions. While using the I-Connect application, Miranda increased her on-task behavior to 90% of the time with a range from 84% to 97% across sessions, and self-recorded her own behavior using the I-Connect application as on-task an average of 97.4% of the time. The range of amount of time on-task recorded by the participant was 93% to 100% of time.

Maintenance probe results. Maintenance probes were administered 2 weeks and 4 weeks after the last session of return to intervention (B2) phase (see Figure 1). Miranda was recorded as on-task 83% of the time during the 2-week maintenance probe. She recorded herself as on-task 93% of intervals using the I-Connect application. When comparing the time-sampling data to the I-Connect data, the researcher and participant matched 75% of the intervals. During the 4-week maintenance probe, Miranda was on-task 96% of the time. She reported that she was on-task 82% of the time using the I-Connect application. When comparing the participant's data to the time-sampling data, 80% of intervals were matched.

Fidelity of Self-Monitoring

Fidelity of self-monitoring data was collected for all phases. During baseline and withdrawal phases, participants could not access the self-monitoring application and evidence of

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Table 2. Teacher and Student Satisfaction Scale Results.

	Rating		
Teacher items	Teacher 1	Teacher 2	Teacher 3
Rate the extent to which I-Connect had a direct impact on the student's productivity in class	5	4	4
Rate the extent to which I-Connect had a direct impact on the student's assignment completion	5	3	5
Rate the extent to which I-Connect had a direct impact on the student's grade	5	3	4
Rate the extent to which I-Connect had a direct impact on the student's accuracy of work in class	5	2	4
	Rating		
Student items	Keith	Brad	Miranda
When using the I-Connect device, do you complete more work in class?	4	4	5
When using the I-Connect device, do you focus more on assignments and lectures?	4	3	5
When using the I-Connect device, do you have better grades?	5	4	5
When using the I-Connect device, do you earn more points on the assignments worked on during class?	4	4	5

Note. Items were scaled using goal-attainment scaling-based metrics with 1 = *much worse than before*, 2 = *somewhat worse than before*, 3 = *same as before*, 4 = *somewhat better than before*, 5 = *much better than before*.

alternate self-monitoring systems was not observed during data collection periods. This resulted in 0% fidelity of self-monitoring in Baseline and Withdrawal phases for all three participants. Fidelity of self-monitoring during intervention and maintenance phases was calculated as a percentage of responsive intervals (defined as student response to the on-task prompt) divided by total application prompts (opportunities to self-monitor). Keith's average fidelity of self-monitoring was high for Intervention (95% of intervals), Return to Intervention (99% of intervals), and Maintenance (100% of intervals). Brad's average fidelity of self-monitoring was similarly high for Intervention (98%), Return to Intervention (99%), and Maintenance (100%) phases. Miranda's average fidelity of self-monitoring was slightly lower for Intervention (91%), but improved during Return to Intervention (94%) and Maintenance (98%) phases.

Teacher Satisfaction Rating Scale Results

Average ratings of teacher satisfaction were positive for all three participating teachers (see Table 2). The classroom teachers for each participant responded to the Teacher Satisfaction Rating Scale. Participating Teacher 1 (total rating: 20/20) indicated the intervention improved all categories of process (productivity) and outcome (grades) for Keith during intervention phases. Teacher 2 (total rating: 12/20) indicated that Brad made improvement in productivity, moderate improvement in assignment completion and grade, but only slight improvement accuracy of work completion. Miranda's teacher (total rating: 17/20) indicated that she demonstrated significant improvement in the areas of productivity, grade, and accuracy of work during intervention phases.

Student Satisfaction Rating Scale Results

Ratings of student satisfaction were collected along with brief interviews from all subjects (see Table 2). Keith completed the Student Satisfaction Rating Scale independently (total rating: 17/20). He indicated that he had the most improvement in his grade when he used the application by rating the question about grades at a "5." He indicated that his performance was improved when he used the I-Connect application in the areas of work completion, ability to focus, and accuracy of class work. Keith identified the advantages of using the I-Connect as, "It helps me with my work and I get good grades." Brad also completed the Student Satisfaction Rating Scale independently (total rating: 15/20), indicating he had made some improvement in the areas of work completion, class grades, and class work accuracy when using the I-Connect application. Brad also indicated that his performance was the same with or without the application in the area of focusing. Somewhat

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paradoxically, he reported that an advantage of using the I-Connect application was that it kept him focused. Miranda completed the Student Satisfaction Rating Scale with support from the first author (total rating: 20/20) who read the questions to her and explained what each number on the Likert-type scale meant. Miranda indicated that she performed a lot better in all of the areas assessed: work completion, ability to focus, grade performance, and class work accuracy.

Discussion

The purpose of the present study was to evaluate (a) the effects of the I-Connect self-monitoring intervention on improving students' on-task behavior using an ABAB withdrawal design, and (b) the degree to which the I-Connect intervention improved teachers' and students' perceptions of students' academic performance. Data collected within the study resulted in improved on-task behavior for all three students during intervention. A functional relationship between the intervention and on-task behavior was supported by immediate deterioration in students' on-task behavior once the intervention was withdrawn. Furthermore, indirect ratings of academic performance and social validity gathered from participants and teachers supported the use of the intervention.

The study extends the intervention literature base in three important ways. First, the intervention adds to a sparse literature base investigating secondary interventions for students with disabilities in high school settings. The intervention resulted in improved on-task behavior for all three students despite differences in categorical disability (specific learning disability, autism, intellectual disability) and instructional setting (general education, self-contained). The intervention's acceptability across settings and teachers was also a strength for school-based professionals seeking efficient and portable interventions. Second, the study offers a small contribution to the broad evidence base supporting self-monitoring interventions while helping to expand the emerging literature base represented by technologically based, self-monitoring interventions. Third, and perhaps most importantly, the improvements in on-task behavior were achieved with an application that allowed for unobtrusive prompts (flashing) rather than the cued buzzers and chimes commonly used in studies of self-monitoring interventions (Gumpel & David, 2000; Rock & Thead, 2007; Wolfe, Heron, & Goddard, 2000).

Teacher responses resulted in strongly positive ratings of improvement in productivity, assignment completion, and grades for all three students with weaker (but still positive) results for accuracy in student work. On this point, the study aligns with prior work conducted by Maag, Reid, and DiGangi (1993) in which older students' work accuracy demonstrated stronger improvements when monitoring accuracy rather than on-task behavior. It is noted that while stronger improvements for on-task behavior were obtained in the current study than in the Maag study, differences in baseline on-task behavior and self-monitoring format (i.e., paper vs. handheld device) preclude more direct comparisons.

Similar data collected from the students resulted in strongly positive ratings across all four categories (productivity, assignment completion, grades, and accuracy). The relatively stronger ratings of work accuracy by students are unsurprising, given the structure of high school classroom settings. Assignments are often graded with enough time between completion and communication of results that appraising accuracy would be difficult. In addition, a more engaged student completing 20 problems rather than 12 problems could be no more accurate yet receive a significantly better grade, resulting in higher grades and a faulty perception by participating students that assignments are more correct rather than more complete.

Limitations

There are three main limitations of the present study. First, data were not collected to investigate generalization in other classrooms not included within the study. Although this is common to intervention research in general (and self-monitoring studies in particular), it is impossible to determine whether the intervention resulted in improved self-monitoring skills in unobserved classrooms or whether awareness of self-monitoring benefits resulted in student implementation of self-monitoring procedures without the application. Second, although cell phones and handheld devices are now commonplace in American high schools, formal and informal district policies regarding acceptable use vary widely. It is unknown at this time whether these technologies will be embraced or rejected at the classroom level. Third, only indirect measures of academic change were collected. Direct measurement of academic output and accuracy is often difficult to collect and measure in a manner that allows for true comparison between intervention phases, but it does add a powerfully important outcome to consider when selecting interventions.

Future Research

This study included reinforcement for participants when meeting predetermined goals of on-task behavior plus acceptable agreement with direct observers for the on-task variable. The use of both as a criterion for reinforcement is rare within the self-monitoring literature and has not yet been explored using handheld devices/applications at the time of this publication. Other researchers successfully using technologically based, self-monitoring systems have either omitted reinforcement within the intervention

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(Crutchfield et al., 2015; Quillivan et al., 2011; Wills & Mason, 2014) or used a study design that resulted in collateral reinforcement from improved on-task behavior due to a preexisting token economy system (Blood et al., 2011; Gulchak, 2008). The variability of reinforcement structures within the self-monitoring literature has been well documented (Fantuzzo & Polite, 1990; Mooney, Ryan, Uhing, Reid, & Epstein, 2005) and calls for further investigation.

Technological innovations hold promise for schools attempting to maximize intervention resources. However, it remains an open question within the literature whether technological or paper-based formats are any different at the level of outcome or treatment acceptability. Given the absence of direct comparisons between formats for either purpose, practitioners may choose to incorporate technology due to its innovative appeal, or alternatively reject it in favor of more traditional (and better supported) formats. However, even if direct comparison of formats within future research were to yield no significant differences for student outcomes, technologically based systems offer unique advantages in data collection and storage. Reduced time spent on paperwork collection and data entry within tiered interventions free time for school-based professionals to meet other student needs—a worthy goal in itself.

Authors' Note

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